

10.1 DC: Practical Circuits & Kirchhoff's Laws

Question Paper

Course	CIEA Level Physics
Section	10. D.C. Circuits
Topic	10.1 DC: Practical Circuits & Kirchhoff's Laws
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

A battery is marked 9.0 V.

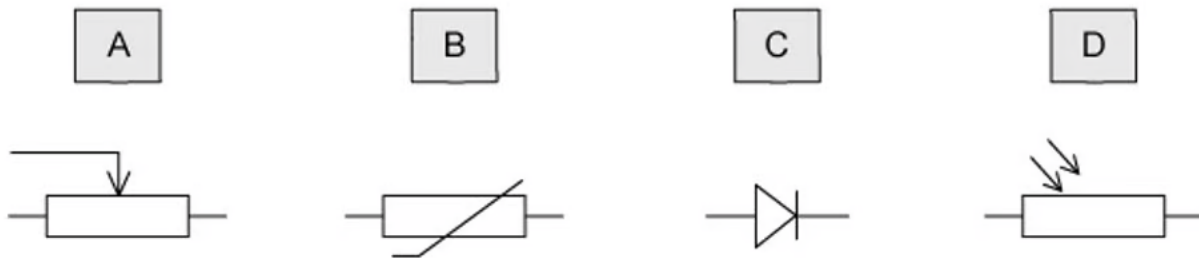
What does this mean?

- A. each coulomb of charge from the battery supplies 9.0 J of electrical energy to the whole circuit
- B. the battery supplies 9.0 J to an external circuit for each coulomb of charge
- C. the potential difference across any component connected to the battery will be 9.0 V
- D. there will always be 9.0 V across the battery terminals

[1 mark]

Question 2

Which symbol represents a component whose resistance is designed to change with temperature?

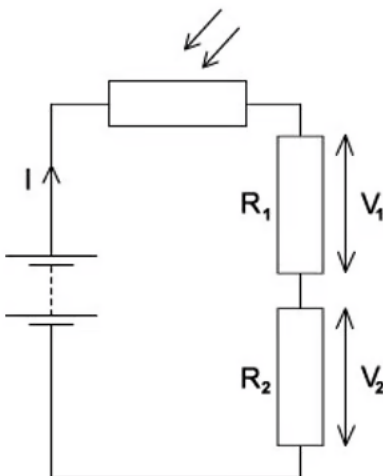


[1 mark]

Question 3

In the circuit shown, a light-dependent resistor (LDR) is connected to two resistors R_1 and R_2 .

The potential difference (p.d.) across R_1 is V_1 and the p.d. across R_2 is V_2 . The current in the circuit is I .



Which statement about this circuit is correct?

- A. the current I increases when the light intensity decreases
- B. the LDR is an ohmic conductor
- C. the p.d. V_2 increases when the light intensity decreases
- D. the ratio $\frac{V_1}{V_2}$ is independent of light intensity

[1 mark]

Question 4

A cell of e.m.f. E delivers a charge Q to an external circuit.

Which statement is correct?

- A. the energy dissipation in the external circuit is EQ
- B. the energy dissipation within the cell is EQ
- C. the external resistance is EQ
- D. the total energy dissipation in the cell and the external circuit is EQ

[1 mark]

Question 5

The principles of conservation of which two quantities are associated with Kirchhoff's first and second Laws?

	First Law	Second Law
A	charge	energy
B	charge	voltage
C	energy	charge
D	voltage	charge

[1 mark]**Question 6**

In deriving a formula for the combined resistance of three different resistors in series, Kirchhoff's Laws are used.

Which physics principle is involved in this derivation?

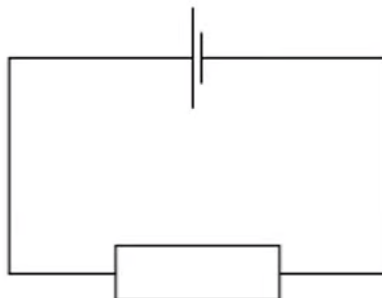
- A. the conservation of charge
- B. the direction of the flow of charge is from negative to positive
- C. the potential difference across each resistor is the same
- D. the current varies in each resistor, in proportion to the resistor value

[1 mark]

Question 7

A cell is connected to a resistor.

At any given moment, the potential difference across the cell is less than its electromotive force.



Which statement explains this ?

- A. the cell is continually discharging
- B. the connecting wire has some resistance
- C. energy is needed to drive charge through the cell
- D. power is used when there is a current in the resistor

[1 mark]

Question 8

The combined resistance R_T of two resistors of resistances R_1 and R_2 connected in parallel is given by the formula

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$$

Which statement is used in the derivation of this formula?

- A. the currents through the two resistors are equal
- B. the potential difference across each resistor is the same
- C. the supply current is split between the two resistors in the same ratio as the ratio of their resistances
- D. the total power dissipated is the sum of the powers dissipated in the two resistors separately

[1 mark]

Question 9

Each of Kirchhoff's two Laws presumes that some quantity is conserved.

Which row states Kirchhoff's **first** Law and names the quantity that is conserved?

	Statement	Quantity
A	The algebraic sum of currents into a junction is zero	charge
B	The algebraic sum of currents into a junction is zero	energy
C	The e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance around the loop	charge
D	The e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance around the loop	energy

[1 mark]**Question 10**

Four statements about potential difference or electromotive force are listed.

- 1 It involves changing electrical energy into other forms
- 2 It involves changing other energy forms into electrical energy
- 3 It is the energy per unit charge to move charge right round a circuit
- 4 It is the work done per unit charge by the charge moving from one point to another

Which statements apply to potential difference and which apply to electromotive force?

	Potential difference	Electromotive force
A	1 and 3	2 and 4
B	1 and 4	2 and 3
C	2 and 3	1 and 4
D	2 and 4	1 and 3

[1 mark]



Model Answers: Easy

1

The correct answer is **A** because a battery being marked 9.0 V means that each coulomb of charge from the battery supplies 9.0 J of electrical energy to the whole circuit (this includes the internal resistance of the battery)

B is incorrect because this would mean that the amount of energy that the battery supplies to an external circuit for each coulomb of charge is not 9.0 V

C is incorrect because if there is no internal resistance in the battery (and the wires have negligible resistance), the sum of p.d. across the components connected to the battery will be 9.0 V

D is incorrect because the e.m.f of a battery does not tell anything about its internal resistance. If the battery has some internal resistance, the p.d. across the battery terminals will not be exactly 9.0 V, but less

2

The correct answer is **B** because:

- A component whose resistance is designed to change with temperature is called a thermistor
- The symbol for a thermistor is given by option **B**

A is incorrect because this is the symbol for a potentiometer

C is incorrect because this is the symbol for a diode

D is incorrect because this is the symbol for an LDR

3

The correct answer is **D** because:

- The resistance of an LDR decreases as the light intensity increases

- The circuit is a series connection, so the same current flows through all the components
- From Ohm's Law or $V = IR$ or $I = \frac{V}{R}$
- Therefore, as the light intensity decreases, the resistance of LDR increases and so current I decreases
- The ratio of V_1 to V_2 is equal to $\frac{V_1}{V_2} = \frac{IR_1}{IR_2} = \frac{R_1}{R_2}$
- The light intensity affects the current in the circuit, but the ratio does not depend on current
- Therefore, the ratio $\frac{V_1}{V_2}$ is independent of the light intensity

A is incorrect because the current I decreases when the light intensity decreases, so light intensity and current are directly proportional

B is incorrect because the LDR is not an ohmic conductor (it does not follow Ohm's Law)

C is incorrect because

since current I decreases as the light intensity on the LDR decreases, which causes the resistance of LDR to increase, $V_2 = IR_2$ will also decrease

4

The correct answer is **D** because:

- The e.m.f. is defined as the total energy provided by the cell per unit charge
- In equation form, this can be written
 - $E = \frac{W}{Q}$
- So, the total energy dissipation in the cell and the external circuit is equal to
 - $W = EQ$

A & B are incorrect because the e.m.f. provides energy across both the cell and the external circuit

C is incorrect because the unit for EQ is equal to the joule J, not the Ohm Ω

5

The correct answer is **A** because:

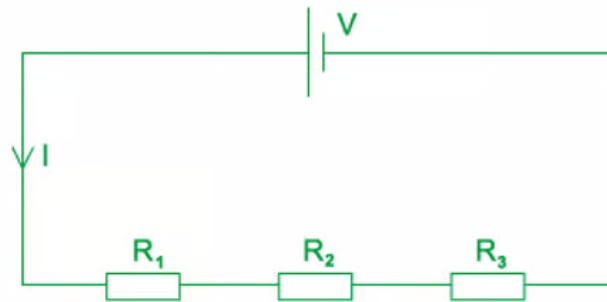
- Kirchhoff's First Law (or current Law) states that current flowing into a junction must be equal to current flowing out of it
 - This is a consequence of charge conservation
 - It is well known in physics that charge will always be conserved. In the context of circuits this means that, since the current is the rate of flow of charge, the current flowing into a point must be the same as the current flowing out of that point
- Kirchhoff's Second Law (or voltage Law) states that the sum of all voltages around any closed loop in a circuit must equal the e.m.f. of the supply
 - This is a consequence of the principle of conservation of energy
 - This is due to the fact that work is done ($W = VQ$) on charge carriers by the electric forces inside the circuit components meaning they either gain or lose electrical energy, depending upon the type of component

6

The correct answer is **A** because:

- The physics principle involved in deriving Kirchhoff's Laws for 3 different resistors connected in series is the conservation of charge
- The shared e.m.f. source V causes a current I to flow through all three resistors, by the conservation of charge, this must be the same at all points in a series circuit
- From Ohm's Law $V = IR$
- The e.m.f. is split into V_1 , V_2 and V_3 between the three resistors
 - $V_1 = IR_1$ $V_2 = IR_2$ $V_3 = IR_3$
- From Kirchhoff's Second Law, we know that the voltages around a loop in a circuit add up to the e.m.f. of the supply
 - $V = V_1 + V_2 + V_3 = IR_1 + IR_2 + IR_3 = I(R_1 + R_2 + R_3)$
 - $\frac{V}{I} = R_1 + R_2 + R_3$
- From Ohm's Law, we know $\frac{V}{I} = \text{total resistance of the circuit}$

○ $R_{total} = R_1 + R_2 + R_3$



B is incorrect because the flow of charge in a circuit is from the positive terminal of the battery to the negative terminal

C is incorrect because in series combination, the p.d. across different resistors is not the same since the resistance of each one is different

D is incorrect because in series combination, the current through each resistor is the same

7

The correct answer is **C** because:

- Most cells tend to have some internal resistance
- The overall e. m. f. provided by the cell is therefore measured as lower than its stated rating
- The e .m .f. of a cell is equal to $E = \frac{W}{Q}$, therefore, energy is needed to drive charge through the cell

A is incorrect because cells discharge over time with continued use, and while this may affect the value of the internal resistance, it is not the reason for the difference between the stated e. m .f. and the measured e. m .f.

B & D are incorrect because the question is in reference to the cell, so the wires and resistor have nothing to do with it

8

The correct answer is **B** because:

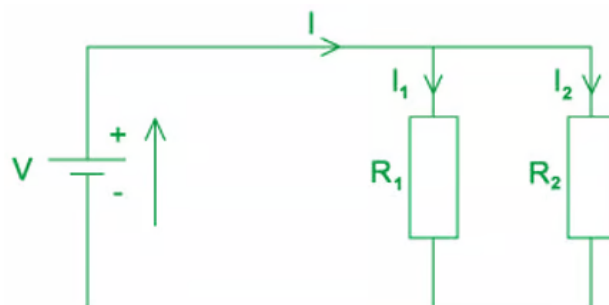
may affect the value of the internal resistance, it is not the reason for the difference between the stated e. m. f. and the measured e. m. f.

B & D are incorrect because the question is in reference to the cell, so the wires and resistor have nothing to do with it

8

The correct answer is **B** because:

- Each resistor in parallel has the same p. d. applied to it, but the total current is split amongst them
- Let the current through resistor R_1 be I_1 and the current through R_2 be I_2
- The p. d. across both R_1 and R_2 is equal to the supply voltage V
- Therefore from Ohm's Law
 - $I_1 = \frac{V}{R_1}$ and $I_2 = \frac{V}{R_2}$
- But from Kirchhoff's Current Law, we know the current entering a junction is equal to the current leaving the junction
 - Therefore, $I = I_1 + I_2$
- Substituting the values derived for I_1 and I_2 gives us'
 - $I = \frac{V}{R_1} + \frac{V}{R_2} = V \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$
- Rearranging gives us
 - $\frac{I}{V} = \frac{1}{R_1} + \frac{1}{R_2}$
- From Ohm's Law, we know $\frac{V}{I} =$ total resistance of the circuit
 - $\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2}$



A is incorrect because in parallel combination, the current through each resistor is different

C is incorrect because the supply current is split between the resistors according to the ratio between the supply p. d. and the resistance of each resistor

$$I = \frac{V}{R_1} + \frac{V}{R_2} = V \left(\frac{R_1 + R_2}{R_1 R_2} \right)$$

D is incorrect because the equation for power is not required in this derivation

9

The correct answer is **A** because:

- Kirchhoff's First Law states: the algebraic sum of currents into a junction is zero
 - This is a consequence of charge conservation

B is incorrect because Kirchhoff's First Law is not a consequence of energy conservation

C & D are incorrect because these two options describe Kirchhoff's Second Law

Exam Tip: Remember Kirchhoff's First and Second Laws like this:

- Kirchhoff's First Law = Current Law = charge conservation ($Q = It$)
- Kirchhoff's Second Law = Voltage Law = energy conservation ($W = VQ$)

10

The correct answer is **B** because:

- Potential difference (p.d.) is defined as the work done per unit charge by the charge moving from one point to another
- It involves changing electrical energy into other forms such as light (bulbs) or sound (speaker)
 - So, statements 1 and 4 refer to p. d.
- Electromotive force (e.m.f.) is defined as the energy per unit charge to move charge right around a circuit

- It involves changing other energy forms into electrical energy, such as chemical energy in the battery
 - So, statements 2 and 3 refer to e.m.f.